

Any fixed number of simultaneous polynomial-size dice

Complete proof; three independent reviews passed

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Status. The full global induction and every input below have passed independent mathematical reviews by the root, size-bound, and rational-design agents. This is not formal verification or a literature-priority claim.

Theorem 1. *For every fixed positive integer k there are constants $C_k, c_k > 0$ such that, for all sufficiently large n , a fully permutation-fair family of n finite equiprobable dice exists with k specified dice having at most $C_k n^{c_k}$ faces each. The remaining $n - k$ dice may have one common finite face count. No bound on that common count is asserted, and no uniform control in growing k is asserted.*

Throughout the proof, polynomial bounds are in n , with constants and exponents allowed to depend on the fixed k . In particular a finite number of polynomial constructions remains polynomial; this convention does not permit taking the least common multiple of independently produced polynomial-sized denominators.

Lemma 1 (One insertion with a reserve of low moments). *Fix $s \geq 1$. Suppose j rational probability densities $f_1, \dots, f_j$ are supported on pure cells of one uniform rational grid of mesh $1/Q$: every open grid cell is occupied by exactly one density. Suppose:*

1. Q , all polynomial piece degrees, and the inverse of every positive cell mass are polynomially bounded;
2. every normalized conditional density, rescaled to $[0, 1]$, lies between fixed positive constants;
3. all raw unnormalized cell moments through degree $2s + 1$ have one polynomial common denominator D .

Then there are $j + 1$ rational densities on a refined collection of pure rational cells, with the same bounds, whose ordered polynomial moments through any prescribed polynomial degree M coincide with those of independent variables having densities $f_1, \dots, f_j, 1$. All new cell endpoints and all new raw cell moments through degree s have one polynomial common denominator. Applying uniform-grid refinement restores the initial uniform-cell form without changing any of these ordered moments or the polynomial bounds.

Proof. For an old cell $I = [A/Q, (A + 1)/Q]$, let γ be its positive mass and w_I its normalized conditional density in the coordinate $t = Qx - A$. Its moments satisfy

$$\mu_{I,a} := \int_0^1 t^a w_I(t) dt = \gamma^{-1} \sum_{l=0}^a \binom{a}{l} Q^l (-A)^{a-l} \int_I x^l f_i(x) dx. \quad (1)$$

Choose one polynomial integer N , divisible by D , sufficiently large that all integers $K_I = N\gamma$ satisfy the count threshold of the synchronized local completion theorem. This is possible because the cell masses are at least $1/D$ and all local complexity bounds are polynomial. For $a \leq 2s + 1$, equation (1) gives

$$K_I \mu_{I,a} = N \sum_{l=0}^a \binom{a}{l} Q^l (-A)^{a-l} \int_I x^l f_i(x) dx \in \mathbb{Z}. \quad (2)$$

Thus the low-moment arithmetic hypothesis of `../size_bounds/synchronized_local_moment_completion.tex` holds simultaneously for the entire polynomial-sized family of local inputs. The degrees and conditional density bounds also satisfy its hypotheses uniformly.

Apply that theorem with output reserve s and ordered exactness M . It supplies one common polynomial denominator L for all low gap moments of the new uniform variable, one common polynomial denominator E for all low *conditional* moments of the old-variable boxes, and one common polynomial endpoint grid. These common denominators are shared by every local instance; they do not result from taking least common multiples after independent constructions. Each old box has normalized mass $1/K_I$.

Rescale the outputs back to I . The old variable's global box mass is $\gamma/K_I = 1/N$, and its raw moment of degree $a \leq s$ is

$$\frac{1}{NQ^a} \sum_{l=0}^a \binom{a}{l} A^{a-l} \nu_l,$$

where ν_l is the normalized box moment with denominator dividing E . All such moments therefore have common denominator dividing $NQ^s E$. The new variable's global mass on the macrocell is $1/Q$; its raw moment on a new local gap is

$$\frac{1}{Q^{a+1}} \sum_{l=0}^a \binom{a}{l} A^{a-l} \rho_l,$$

where the local unnormalized gap moments ρ_l have denominator dividing L . Their common denominator divides $Q^{s+1}L$. All global endpoints have the common local grid denominator multiplied by Q . Multiplying this fixed number of common denominators gives a single polynomial denominator for the new global cells and their low moments. The quantitative local construction gives inverse-polynomial cell lengths, controlled conditional density bounds, and polynomial piece degrees. Positive masses are at least the reciprocal of the common denominator.

Assemble the old outputs by their existing labels and the new uniform outputs as label $j+1$. The resulting densities have total mass one. To check ordered moments, partition their product domain according to original grid cells. The j old variables occupy distinct cells because the cells were pure. The new variable can therefore share a cell with at most one old variable. If it shares none, only marginal polynomial moments are needed; if it shares one, the two-variable ordered local moment identity applies in that cell. All relative orders between other cells are fixed. Expand a degree- M test polynomial into its monomials and apply these local identities and the unchanged cell masses. Summing proves every required global ordered moment identity.

Finally apply `../size_bounds/uniform_grid_moment_refinement.tex` with low reserve s and high degree M . Its input assumptions hold: there is one polynomial endpoint grid, a common polynomial denominator for the low moments, inverse-polynomial cell lengths, and polynomial upper and reciprocal lower density bounds. The latter follow from the conditional bounds and the polynomial mass/length bounds. The refinement preserves every original cell moment through degree M , retains pure labels and the low common denominator, and makes the ambient cells uniform. Thus it preserves all ordered degree- M moments and finishes the lemma. $\square$

Proof of Theorem 1. Fix k . Put

$$q_j = 2^{k-j+1} - 1 \quad (1 \leq j \leq k), \quad q_j = 2q_{j+1} + 1.$$

These are fixed integers, with $q_k = 1$. Work throughout with ordered polynomial exactness $M = n + q_1$; using this larger degree is harmless. Begin with the single density $f_1 = 1$ on the one-cell grid. It has all required conditional bounds, and its moments through q_1 have a fixed common denominator. Together with $n - 1$ independent uniforms it is fully permutation-fair.

Inductively, after constructing j special densities, suppose their pure cells lie on a uniform polynomial grid and their raw moments through degree q_j have one polynomial common denominator. Apply Lemma 1 with $s = q_{j+1}$ to separate one of the remaining uniforms as the next special density. The reserve identity gives exactly the needed input moments. The lemma preserves every ordered polynomial moment through degree M of the $j + 1$ special variables and restores the same invariant with reserve q_{j+1} . In a fixed full order of all n variables, integration over the remaining $n - j - 1$ uniforms leaves, on each special-variable chamber, a polynomial of degree $n - j - 1 \leq M$. Hence full permutation fairness is preserved. There are only $k - 1$ insertions, so every complexity and common denominator bound remains polynomial in n , with exponents depending on k .

After the last step there are k rational pure densities fully fair with $n - k$ uniforms. Their cell masses have a common polynomial denominator D_* . Choose a common polynomial factor H so that on each cell of mass a/D_* there is a real equal-weight quadrature with exactly Ha distinct interior nodes, exact through the larger of degree n and one more than the polynomial degree of that cell density. The separated quantile Newton lemma in `../size_bounds/generalized_local_two_variable_completion.tex` applies uniformly because all conditional density bounds and polynomial degrees are controlled. Increase H also so that $H \geq (n - k)^2$. Each special variable now has exactly HD_* equiprobable nodes; all k specified counts may therefore be chosen equal and polynomial.

On each product of pure cells the order is fixed, so the unchanged cell moments preserve every full-order kernel. Each cell's discrete cluster has at least $(n - k)^2$ distinct interior nodes and rational moments through degree $n - k$. Shrink the finitely many cells around their nodes to obtain disjoint cluster closures. The conditional lifting theorem in `../size_bounds/clustered_multi_exceptional_lifting.tex` then converts the remaining uniforms into finite dice, retaining every specified count HD_* . It provides one common finite count for those remaining dice. Globally order the face labels and replace them by successive integers to make all labels distinct. This proves the theorem. $\square$